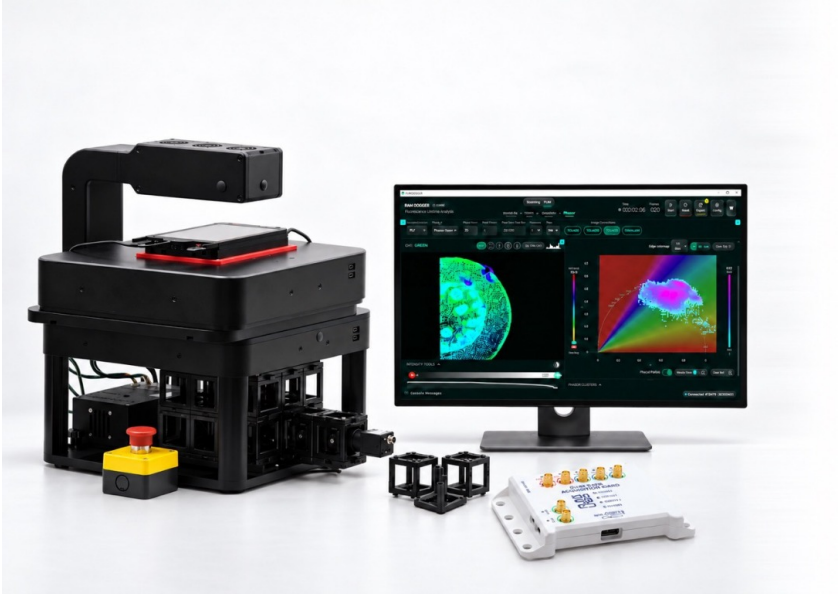
User Manual

# openUC2 FLIM Upgrade Kit

**MU-UC2-01-1C1L**



User Manual – Version v1.1.0
30 Oct 2026

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# 1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, regulatory compliance, warranty conditions, and the conventions used throughout this document.

## 1.1 About This Manual

This manual provides instructions for the installation, configuration, operation, maintenance, and disposal of the openUC2 FLIM Upgrade Kit. The upgrade kit enables fluorescence lifetime imaging (FLIM) measurements on compatible openUC2 FRAME microscope systems by integrating FLIM LABS hardware, optical components, accessories, and software. Depending on the ordered configuration, the supplied components may vary. Individual components may be supplied with their own dedicated manuals containing additional safety, installation, and operating information. These manuals form part of the product documentation and must also be read and followed.

The microscope system itself is **not** part of the upgrade kit. The user is responsible for the installation, operation, and maintenance of the connected openUC2 FRAME according to the documentation provided by openUC2 and all applicable safety instructions.

The upgrade kit has been developed and validated for compatibility with supported openUC2 FRAME configurations in collaboration with openUC2. FLIM LABS S.r.l. is an independent manufacturer and is not affiliated with openUC2.

This manual forms an integral part of the product and must remain accessible to all personnel responsible for installation, operation, maintenance, and servicing. If the product is transferred to a third party, all accompanying documentation must also be transferred.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.



**Warning** Before performing any operation, read this manual together with the documentation of the connected microscope system and all supplied components.

### 1.1.1 User Qualification

The upgrade kit is intended for use by trained laboratory personnel familiar with laboratory electronic instrumentation and optical equipment. Personnel shall have received appropriate instruction regarding the installation, operation, maintenance, and safety precautions described in this manual.

The upgrade kit is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may operate the upgrade kit only under the supervision of qualified personnel. Installation and operation require familiarity with the connected microscope system and the associated manufacturer documentation. The user is responsible for ensuring that all microscope-related safety instructions and operational limitations specified by the microscope manufacturer are followed.

### 1.1.2 Revision History

Document Revision: v1.1.0 - 30 Oct 2026

A certified copy of this manual is stored in the manufacturer's technical file for the legally required period. During this period, documentation may be requested from the manufacturer.

## 1.2 Symbols and Conventions

This manual includes two different levels of hazard warnings. The warnings may include various safety symbols. The hazard levels and safety symbols comply with the ISO Standard for Product Safety Signs and Labels.

### Hazard Levels



**Warning** Indicates a hazardous situation that, if not avoided, could result in death or serious injury.

**Caution** Indicates a hazardous situation that, if not avoided, could result in minor or moderate injury.

**Notice** Indicates important information intended to prevent equipment damage, malfunction, or improper operation.

### Safety Symbols

Table 1.1 shows a list of symbol types that can be found on the product or in the manual.

**Table 1.1:** Safety symbols and their meanings

| Symbol | Form | Information |
|---|---|---|
|  | Yellow triangle with a black border and a black symbol | Indicates a hazard. |
|  | Circular sign with a red border, diagonal line, and black symbol | Identifies a prohibited action. |
|  | Blue circle with a white symbol | Identifies a required action. |

**Warning** Safety pictograms performing a protective function must not be removed, covered, or damaged. If a label becomes illegible or damaged, contact authorized service for replacement.

## 1.3 Manufacturer Information

Manufacturer: FLIM LABS S.r.l.
Address: Via della Farnesina, 3 00135 Rome (RM), Italy
Telephone: +39-329 761540
E-mail: info@flimlabs.com
Website: www.flimlabs.com

## 1.4 Regulatory Compliance

The FLIM LABS components supplied with the upgrade kit are individually CE-marked where applicable and are accompanied by their respective EU Declarations of Conformity in accordance with the applicable European legislation.

The connected microscope system is an independent third-party product manufactured and certified by its respective manufacturer. FLIM LABS S.r.l. does not certify or modify the regulatory compliance of the microscope system.

The upgrade kit has been tested for compatibility with supported openUC2 FRAME configurations. Compatibility testing does not constitute certification or regulatory approval of the complete integrated microscope system.

The user is responsible for ensuring that the complete installed system complies with all applicable regulations, safety requirements, and operating conditions after installation.

## 1.5 Product and Component Labels

The components supplied with the product contain labels with regulatory markings and identification information including:

- Product name and model

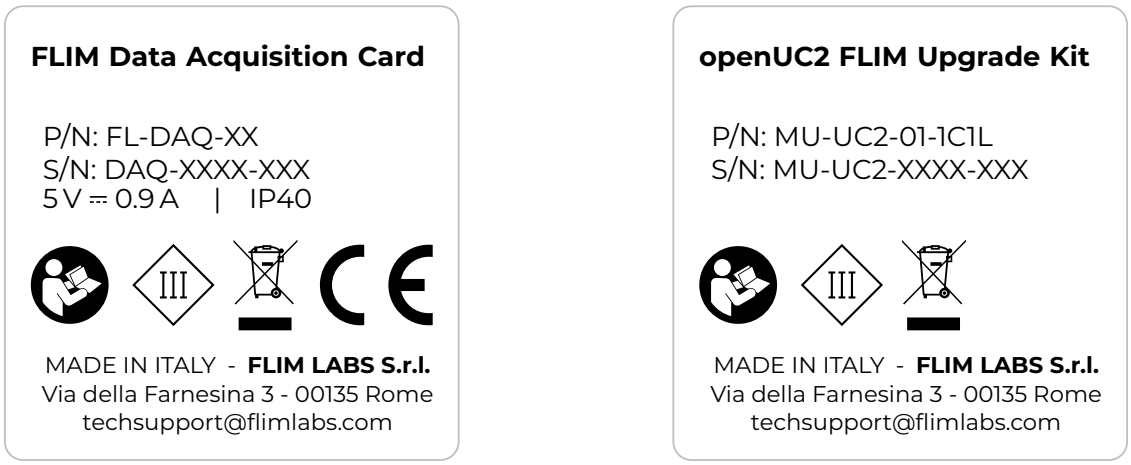



**Figure 1.1:** Product labels for the kit components

- Manufacturer identification
- Regulatory markings where applicable
- Electrical specifications
- Serial number information

The product packaging may additionally contain kit-level identification information including the product number and configuration information.

Disposal requirements associated with WEEE markings are described in Section 7.2.

## 1.6 Warranty and Liability

### Warranty

The manufacturer provides 24 months warranty for private customers and 12 months warranty for commercial customers. The warranty period begins on the date of delivery.

The warranty covers defects in materials and workmanship under normal and intended use. It is valid only if the product is installed, operated, and maintained according to this manual and the applicable manuals of the supplied components.

Unauthorized modification, improper use, failure to follow instructions, or repair by non-authorized personnel voids the warranty.

### Limitation of Liability

The FLIM LABS components supplied with the upgrade kit have been designed, manufactured, and tested in accordance with applicable European directives and internal quality standards.

The upgrade kit is intended for integration with supported microscope configurations. However, the complete microscope system remains a third-party system outside the control of FLIM LABS S.r.l.

The manufacturer shall not be liable for damages resulting from improper use, including but not limited to:

- Use outside the intended purpose
- Failure to follow this manual or applicable safety instructions
- Failure to follow the documentation of the microscope manufacturer
- Installation or operation by unqualified personnel
- Operation outside specified electrical, optical, or environmental limits
- Use of incompatible or non-compliant input signals or connected equipment
- Unauthorized modification, repair, or use of non-original spare parts
- Integration into systems not compliant with applicable safety regulations

The user is responsible for ensuring that the complete integrated system is installed, operated, and maintained in accordance with all applicable local, national, and international regulations.

Except where otherwise required by mandatory law, the manufacturer shall not be liable for indirect, incidental, or consequential damages, including but not limited to loss of data, loss of profit, or system downtime.

# 2 System Description

This chapter provides an overview of the openUC2 FLIM Upgrade Kit, its intended application, major system components, and the overall system architecture.

The upgrade kit extends compatible openUC2 FRAME microscope systems with fluorescence lifetime imaging (FLIM) capabilities using FLIM LABS timing electronics, pulsed laser excitation, and single-photon detectors.

The kit is intended as an integration solution for supported openUC2 FRAME microscope configurations and does not replace or modify the primary functionality of the microscope system.

## 2.1 Functional Description

The openUC2 FLIM Upgrade Kit enables time-resolved fluorescence measurements on compatible openUC2 FRAME microscope systems.

The system operates by exciting the sample with a pulsed laser source synchronized to the *FLIM DAQ*. Fluorescence photons emitted by the sample are detected by a FLIM LABS SPAD detector and recorded using time-correlated single-photon counting (TCSPC).

The openUC2 FRAME scanning system provides pixel, line, and frame synchronization signals required for image reconstruction. These synchronization signals are connected directly to the *FLIM DAQ*.

The acquired photon timing information is processed by *FLIM IMAGER* to generate fluorescence lifetime images and related analysis data.

The upgrade kit interfaces with the microscope without modifying its primary functionality. The openUC2 FRAME microscope remains responsible for optical imaging and sample scanning, while the FLIM LABS hardware provides pulsed laser excitation, single-photon detection, photon timing acquisition, and fluorescence lifetime analysis.

## 2.2 Intended Use

The openUC2 FLIM Upgrade Kit is intended for laboratory and research applications involving fluorescence lifetime imaging microscopy (FLIM).

The system is intended for integration with supported openUC2 FRAME configurations by qualified personnel familiar with microscopy systems, laser-based instrumentation, and laboratory electronics.

The upgrade kit is intended exclusively for indoor laboratory use under the environmental and operating conditions specified in this manual and in the manuals of the supplied components.

The upgrade kit is not intended for:

- Medical diagnosis or therapeutic applications
- Operation in explosive or hazardous environments
- Outdoor operation
- Safety-critical applications
- Use with unsupported microscope configurations

## 2.3 Reasonably Foreseeable Misuse

Any use outside the intended purpose described in this manual is considered improper use.

Reasonably foreseeable misuse includes, but is not limited to:

- Operation with unsupported microscope systems
- Modification of the supplied hardware or optical components
- Use of incompatible electrical or optical signals
- Bypassing safety mechanisms of the microscope or laser system
- Simultaneous emission from multiple excitation lasers or combining the outputs of multiple laser sources into a common optical fiber or optical path, unless explicitly permitted by the manufacturer.

- Operation outside specified environmental conditions
- Use without reading the supplied documentation
- Use by unqualified personnel

Improper use may result in equipment damage, incorrect measurements, reduced performance, or hazardous operating conditions.

## 2.4 System Components Overview

The standard openUC2 FLIM Upgrade Kit contains the components listed in Table 2.1. The number of components and the specific product number depend on the ordered upgrade configuration. Product variants are indicated with an *X* in the product number.

**Table 2.1:** System components overview

| Component | FLIM LABS Part P/N | Description |
|---|---|---|
| Picosecond Pulsed Fiber-Coupled Laser | PLD-XXX-SM-FC-U | Excitation source for fluorescence lifetime measurements. Different excitation wavelengths are available depending on the ordered configuration. |
| Single-Photon SPAD Detector | SPAD-050-C-X-FC | Single-photon avalanche diode (SPAD) detector used for fluorescence photon detection. |
| *FLIM DAQ* | FL-DAQ-0X | Two-channel time-correlated single-photon counting (TCSPC) acquisition hardware used for photon timing measurements and synchronization signal acquisition. |
| Filtered multimode optical fiber assembly | n.a. | Provides fluorescence collection and spectral filtering before the fluorescence signal reaches the SPAD detector. |
| Single-mode laser fiber | n.a. | Single-mode optical fiber used to deliver excitation light from the pulsed laser to the microscope. |
| Imaging connection cables | n.a. | Electrical cables used to interface the microscope synchronization signals with the FLIM LABS acquisition hardware. |

## 2.5 System Architecture

Figure 2.1 shows the overall architecture of the openUC2 FLIM Upgrade Kit. The system combines the openUC2 FRAME microscope platform with FLIM LABS timing electronics, pulsed laser excitation, and single-photon detection.

The pulsed laser source is connected to the microscope using the supplied single-mode optical fiber. Fluorescence emission generated within the sample is routed through the supplied filtered multimode optical fiber assembly before reaching the SPAD detector.

The openUC2 FRAME scanning system provides pixel, line, and frame synchronization signals required for FLIM image reconstruction. These synchronization signals are connected directly to the *FLIM DAQ*.

The laser synchronization output is connected directly to the *SYNC-IN* input of the *FLIM DAQ*.

The *FLIM DAQ* records

- Photon timing signals from the SPAD detector
- Pixel synchronization signals
- Line synchronization signals
- Frame synchronization signals

The acquired timing information is transferred to a connected computer through USB and processed by *FLIM IMAGER* software for fluorescence lifetime analysis and image reconstruction.
The openUC2 FRAME microscope remains responsible for optical imaging and sample scanning, while the FLIM LABS hardware performs excitation, photon timing acquisition, and fluorescence lifetime analysis.

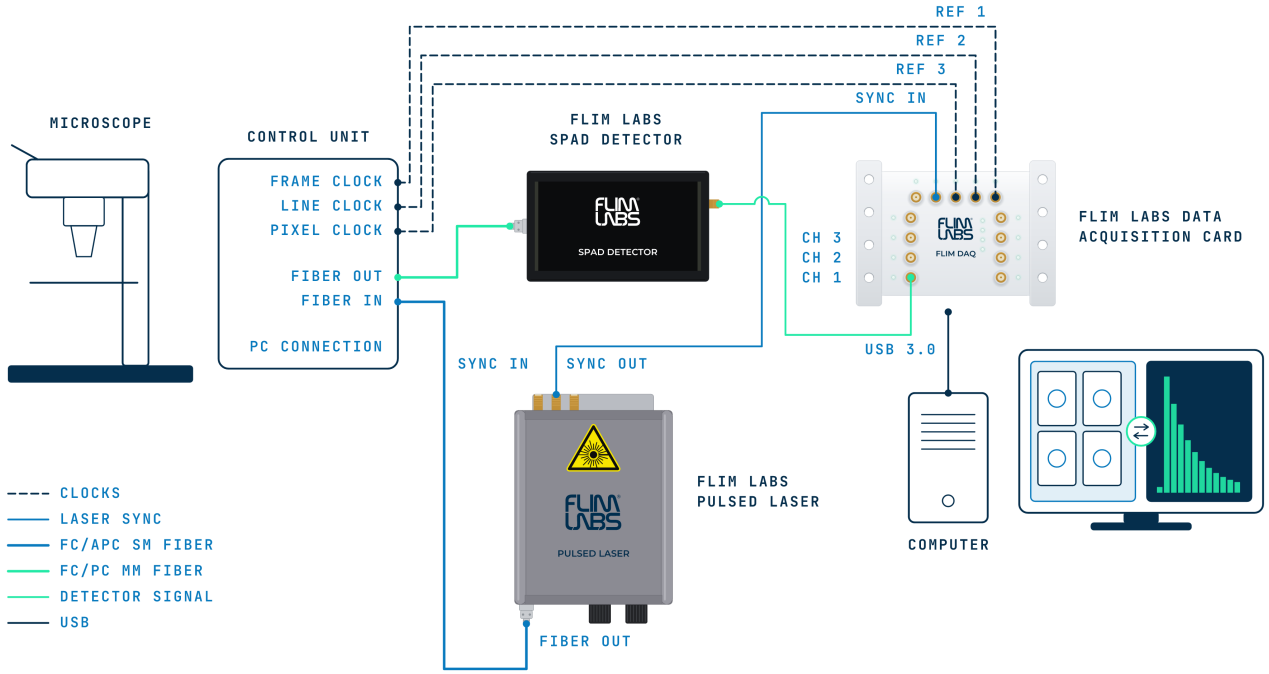



**Figure 2.1:** System architecture of the openUC2 FLIM Upgrade Kit.

## 2.6 System Requirements and Validated Configurations

The openUC2 FLIM Upgrade Kit is intended for use with compatible openUC2 FRAME microscope systems that have been validated by FLIM LABS S.r.l.
Validation may depend on factors including:

- openUC2 FRAME microscope model and configuration
- Detector configuration
- Laser wavelength
- Installed microscope options
- Software and firmware versions

The microscope system itself is not part of the upgrade kit and must be supplied separately by the user.
The upgrade kit has been developed and validated for compatibility with supported openUC2 FRAME configurations in collaboration with openUC2. FLIM LABS S.r.l. is an independent manufacturer and is not affiliated with openUC2. Compatibility with configurations not validated by FLIM LABS S.r.l. is not guaranteed.

**Table 2.2:** Validated configurations

| Microscope | Detector | Configuration | Status |
|---|---|---|---|
| UC2 FRAME Microscope | FLIM LABS SPAD Detector | Standard upgrade kit configuration | Validated |

### System Requirements

The *FLIM DAQ* requires a compatible Windows computer for configuration and operation. The computer shall satisfy the minimum requirements of the *FLIM IMAGER* software:

- PC with multicore CPU
- 8 GB RAM
- USB interface for connecting the DAQ to the PC
- Windows 10 or later

Refer to the latest software documentation for updated system requirements.

# 3 Safety Information

This chapter describes general safety instructions and residual risks associated with the openUC2 FLIM Upgrade Kit and its integration into a microscope system.

The complete integrated system may contain additional hazards not originating from the FLIM LABS upgrade kit itself, including laser radiation, moving mechanical assemblies, motorized stages, optical radiation, electrical hazards, heated components, and other configuration-dependent risks.

The user is responsible for reviewing and following all safety instructions provided by the microscope manufacturer and the manufacturers of all connected accessories and subsystems.

## 3.1 General Safety Instructions

Before installation and operation, ensure that this manual and the documentation of the connected microscope system are complete and legible. If any part of the documentation is missing or unclear, contact the respective manufacturer before proceeding.

The upgrade kit must be used only as described in this manual and only with supported microscope configurations. Any operation not explicitly described in this manual is considered improper use.

This manual complements applicable national and local safety regulations. If local regulations impose stricter requirements, those regulations take precedence.

Despite the implemented safety measures, residual risks may remain if the system is not installed, operated, or maintained as specified.

The microscope itself is not part of the upgrade kit and remains an independent third-party product manufactured and certified by its respective manufacturer. The user remains responsible for following all safety instructions and operational limitations specified by the microscope manufacturer.



**Warning** Before performing any installation or electrical connection procedure, disconnect power from the microscope system and all connected FLIM LABS devices unless explicitly instructed otherwise.



**Warning** Do not modify or bypass microscope safety interlocks, protective covers, or safety mechanisms during installation or operation of the upgrade kit.

## 3.2 Laser Safety

Depending on the selected configuration and the microscope system used, the complete system may contain one or more visible and/or invisible laser sources capable of causing eye injury.

Follow all laser safety instructions provided by the microscope manufacturer and the documentation supplied with the FLIM LABS laser. The supplied laser is accompanied by its own product manual, safety labels, and regulatory information, which form part of the complete system documentation.



**Caution** Visible and invisible laser radiation.
Avoid direct eye exposure to laser radiation.
Do not disconnect optical fibers while the laser source is active.
Do not modify optical safety components or microscope safety interlocks.
Operate the system only under safe operating conditions and according to all supplied documentation.

## 3.3 Electrical Safety

The FLIM LABS upgrade kit contains electronic devices intended for laboratory use.

- Operate the FLIM LABS components only with the specified power supplies and interfaces.
- Use only undamaged and properly rated cables.
- Disconnect power before modifying electrical or synchronization signal connections.
- Do not open the enclosures of FLIM LABS electronic devices.

- There are no user-serviceable internal components.
- Keep all electronic components away from water sources and other liquids.
- Ensure that cables are routed to avoid mechanical damage and accidental disconnection.
- Unauthorized repair or modification is not permitted.

The connected microscope system may contain additional electrical hazards and power supplies not covered by this manual. The user must follow all electrical safety instructions provided by the microscope manufacturer.

## 3.4 Environmental and Operating Conditions

The product is designed for indoor laboratory use only.

Operate the supplied components only within the environmental limits specified in the respective component manuals.

Typical operating conditions include:

- Indoor laboratory environment
- Stable, vibration-free installation surface
- Non-condensing humidity conditions
- Adequate ventilation around electronic components
- Protection from liquids, dust, and mechanical impact

Do not operate the system in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment procedures to avoid connection errors or accidental damage to optical components and connectors.

The connected microscope system may have additional environmental limitations specified by the microscope manufacturer.

## 3.5 Third-Party System Safety

The openUC2 FLIM Upgrade Kit is intended for integration into a third-party microscope system.

Depending on the installed microscope configuration, the complete integrated system may contain additional hazards including:

- Additional laser sources
- Motorized positioning stages
- Moving optical assemblies
- High-intensity optical radiation
- Heated components
- Immersion liquids
- Sharp optical or mechanical components

These hazards are not controlled by FLIM LABS S.r.l. and may vary depending on the microscope configuration and installed accessories.

The user is responsible for:

- Following all safety instructions provided by the microscope manufacturer
- Verifying safe operation of the integrated system
- Ensuring compatibility of all connected devices and accessories
- Maintaining access to all required safety documentation

## 3.6 Modification and Improper Use

The FLIM LABS components supplied with the upgrade kit must not be modified, altered, or repaired by unauthorized personnel.

Unauthorized modification of:

- Electrical connections

- Optical connections
- Laser safety features
- Microscope safety mechanisms
- Signal conditioning electronics

may result in hazardous operating conditions, incorrect measurements, equipment damage, or loss of regulatory compliance.

Use outside the intended application, operation beyond specified limits, or failure to follow the procedures described in this manual and in the microscope documentation may result in malfunction and void the warranty.

Only validated system configurations approved by FLIM LABS S.r.l. should be used.

# 4 Setup

This chapter describes the correct procedure for unpacking, installing, connecting, and verifying the openUC2 FLIM Upgrade Kit before operation.

The upgrade kit is intended for integration into compatible openUC2 FRAME microscope systems by qualified personnel familiar with laboratory instrumentation, microscopy systems, and laser-based equipment.

## 4.1 Scope of Delivery

The openUC2 FLIM Upgrade Kit is supplied in a single standard configuration. Table 4.1 lists the components included with the upgrade kit.

The openUC2 FRAME microscope platform itself is not part of the upgrade kit and must be supplied separately by the user.

**Table 4.1:** Kit configuration and scope of delivery

| Component | MU-UC2-01-1C1L |
|---|---|
| Picosecond Pulsed Laser | 1 |
| Single-Photon SPAD Detector | 1 |
| *FLIM DAQ* | 1 |
| Filtered multimode optical fiber assembly | 1 |
| Single-mode laser fiber | 1 |
| Electrical connection cable set | 1 |
| USB cables and power supplies | 1 set |

Verify that all required items are present before proceeding with installation.

## 4.2 Unpacking and Inspection

After unpacking, carefully inspect all components and accessories:

- Check for visible damage.
- Check for dents, deformation, cracks, or signs of impact.
- Verify that all connectors and cables are intact.
- Inspect optical connectors and optical surfaces for contamination or damage.
- Verify that all ordered accessories are present.

If any damage, missing parts, or abnormalities are detected, stop the installation and contact the manufacturer before proceeding.



**Warning** Keep packaging materials out of reach of children and dispose of them according to local regulations.

## 4.3 Installation Requirements

Before installing the upgrade kit, ensure that the safety requirements described in Chapter 3 are fulfilled, including the specified environmental and operating conditions.

Verify the following before installation:

- The openUC2 FRAME microscope is installed and operational according to the documentation provided by openUC2.
- The microscope provides the required synchronization signals for the selected upgrade configuration.
- A compatible Windows computer with an available USB interface is available for the *FLIM DAQ*.

The computer shall satisfy the minimum system requirements of the *FLIM IMAGER* software as described in Section 2.6.

## 4.4 Hardware Installation

All installation procedures and electrical connections must be performed by qualified personnel familiar with laboratory optical instrumentation and the documentation of all connected devices.

**Warning** Before installation, switch off the microscope system and all FLIM LABS components unless explicitly instructed otherwise.
Do not modify microscope safety covers, laser interlocks, or protective enclosures during installation of the upgrade kit.

### 4.4.1 Mechanical Installation

Install the supplied FLIM LABS components in suitable locations that allow safe operation and convenient access.

Ensure that:

- All components are securely supported.
- Cables and optical fibers are protected from mechanical stress.
- Optical fibers are not bent below their minimum bending radius.

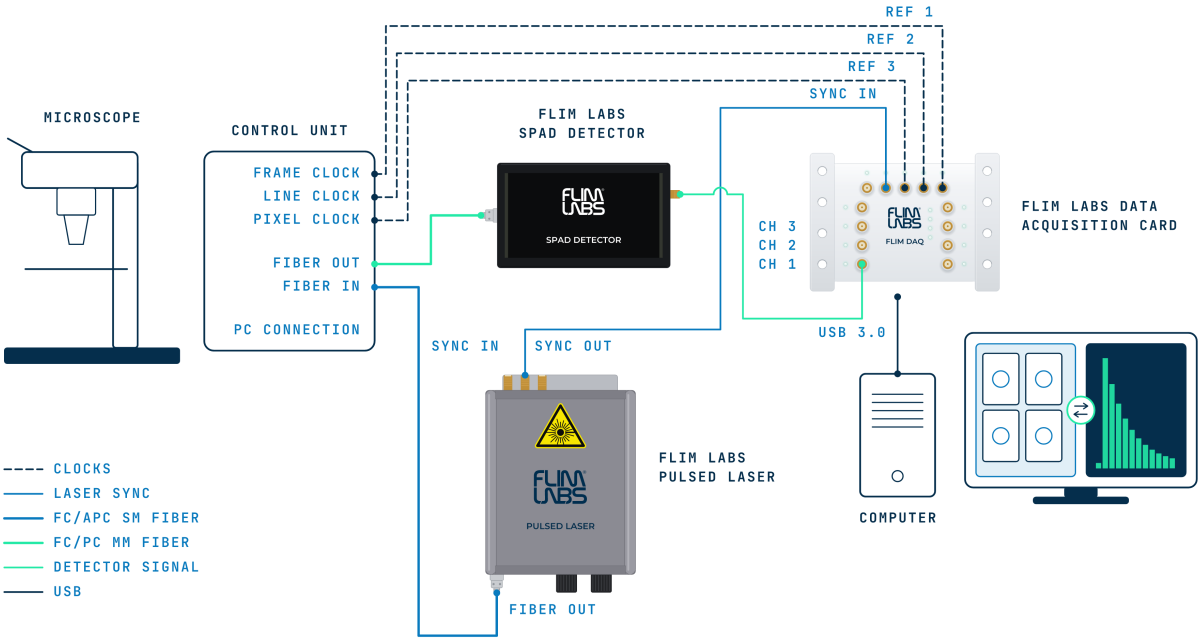



**Figure 4.1:** Connection overview of the openUC2 FLIM Upgrade Kit.

### 4.4.2 Optical Connections

This kit uses optical fibers to transport laser and fluorescence signals. Before connecting optical fibers ensure that:

- Optical connectors are clean before connection.
- Fiber connectors are correctly aligned before tightening.
- Optical fibers are protected from excessive bending and mechanical stress.
- Dust caps are removed only immediately before connection.

Ensure that the laser emission is disabled before connecting or disconnecting any optical fiber

**Pulsed Laser Source**

Connect the FC/PC optical output port of the FLIM LABS pulsed laser to the FC/PC optical input port of the microscope excitation path using the supplied single-mode optical fiber.

**SPAD Detector**

Connect the FC/PC optical output port of the microscope fluorescence detection path to the FC/PC optical input port of the SPAD detector using the supplied filtered multimode optical fiber assembly. Ensure that the filter assembly is oriented according to its markings so that fluorescence passes through the filter before reaching the detector.

Verify that all optical connections are secure and that the fibers are free from excessive bending or mechanical stress before enabling laser emission.

### 4.4.3 Electrical Connections

The system connections are shown in Figure 4.1.

**Detector Signal**

Connect the SPAD detector output directly to the detector input of the *FLIM DAQ*.

**Synchronization Signals**

The openUC2 FRAME controller provides pixel, line, and frame synchronization signals required for FLIM image reconstruction.

Connect these synchronization signals directly to the corresponding reference inputs of the *FLIM DAQ*.

Connect the *Sync-OUT* output of the FLIM LABS laser directly to the *SYNC-IN* input of the *FLIM DAQ*.

**Data Connection**

Connect the *FLIM DAQ* to the acquisition computer using the supplied USB cable.

### 4.4.4 Power Supply

Connect all devices only to compatible power supplies and USB interfaces.

Ensure that:

- Power supplies are correctly rated.
- USB cables are securely connected.
- Cables are routed safely and protected from mechanical damage.

Do not apply power to the system until all optical and electrical connections have been completed and verified.

## 4.5 Software Installation

Before installation verify that the connected computer satisfies the system requirements described in Section 4.3.

Data acquisition and fluorescence lifetime analysis is performed using the *FLIM IMAGER* software. The software can be downloaded from the FLIM LABS website:
https://www.flimlabs.com/products/FLIM-imager.

The download contains a Windows installer file which includes the drivers required for operation of the *FLIM DAQ*. Follow the on-screen instructions for installation on your PC.

## 4.6 Initial Verification

After completing the hardware installation and software installation:

1. Power on the openUC2 FRAME microscope system according to the microscope manufacturer's instructions.
2. Power on all FLIM LABS components.
3. Start the *FLIM IMAGER* software.
4. Select *Guided Configuration* to start the configuration wizard.
5. Verify that the *FLIM DAQ* is detected.
6. Detect or select the active acquisition channels.
7. Select the appropriate timing synchronization source (typically *SYNC-IN* when using the supplied FLIM LABS laser).

8. Detect or configure the reference signals from the microscope scanning system.
9. Configure the image dimensions and scan area based on your preference.
10. Use the live preview or diagnostic tools to verify that detector and synchronization signals are present.
11. Save the configuration for future measurements.

For detailed software operation, configuration options, acquisition modes, data export, and analysis functions, refer to the dedicated *FLIM IMAGER* software manual available on the website.

# 5 Operation

This chapter describes the typical workflow for operating the openUC2 FLIM Upgrade Kit together with the connected openUC2 FRAME and the *FLIM IMAGER* software.

The microscope system and the FLIM acquisition system operate as two coordinated but independent systems:

- The openUC2 control software manages microscope scanning, imaging parameters, and microscope hardware.
- The *FLIM IMAGER* software performs photon acquisition, image reconstruction, fluorescence lifetime analysis, and data export.

Correct operation requires synchronization between the microscope scanning system and the FLIM LABS acquisition hardware.

## 5.1 System Operation Principle

During microscope scanning, synchronization signals generated by the openUC2 FRAME are transmitted to the *FLIM DAQ* card. The detectors measure fluorescence photons generated by the pulsed excitation laser. The *FLIM IMAGER* software reconstructs the fluorescence image using:

- detector photon signals,
- laser synchronization signals,
- microscope pixel, line, and frame synchronization signals.

The reconstructed fluorescence intensity and lifetime information are displayed in real time by the *FLIM IMAGER* software.

## 5.2 Startup Procedure

A typical startup workflow is listed below:

1. Verify that all optical and electrical connections are complete.
2. Power on the microscope system according to the microscope manufacturer's instructions.
3. Power on all FLIM LABS hardware components.
4. Verify that only one excitation laser is enabled to emit at any time. If multiple excitation lasers are installed, ensure that all other laser sources remain disabled.
5. Start the openUC2 control software.
6. Start the *FLIM IMAGER* software.
7. Load the previously saved FLIM LABS configuration.
8. Verify that the FLIM Data Acquisition Card is detected correctly.
9. Verify that detector and synchronization signals are present.
10. Start microscope scanning using the openUC2 control.
11. Start FLIM acquisition in the *FLIM IMAGER* software.



**Notice** The microscope scanning system must be active for correct FLIM image reconstruction. Missing synchronization signals may result in missing or distorted images.

## 5.3 FLIM Acquisition Workflow

The *FLIM IMAGER* software provides several acquisition and visualization modes for fluorescence lifetime imaging. For normal system alignment and signal optimization, *Scouting Mode* is recommended. This mode provides fast intensity-based visualization for convenient setup adjustment.

For fluorescence lifetime analysis, the software provides several acquisition modes including:

- InstaFLIM

- TCSPC
- Calibration
- Phasors

Detailed descriptions of all software modes and analysis functions are provided in the dedicated *FLIM IMAGER* software manual.

During operation, users typically monitor both the openUC2 control and the *FLIM IMAGER* software simultaneously.

For convenient operation, the *FLIM IMAGER* software provides a floating mini-control window that can be opened using the W button. This window allows quick start and stop control of FLIM acquisition while monitoring the host instrument software.

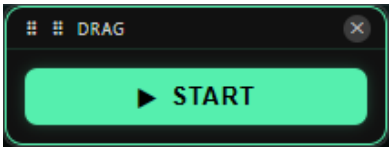



**Figure 5.1:** Floating mini-control window for convenient acquisition control.

## 5.4 Image Reconstruction Modes

The *FLIM IMAGER* software supports several image reconstruction methods depending on the available synchronization signals from the microscope system.

If all trigger signals are present, the recommended reconstruction method is **PLF (Pixel–Line–Frame)**. This mode uses pixel, line, and frame synchronization signals to achieve accurate image reconstruction and spatial alignment.

If synchronization signals are incomplete or unavailable, simplified reconstruction modes may be used for diagnostics or preliminary measurements. Refer to the *FLIM IMAGER* software manual for detailed information.

## 5.5 Data Export

The *FLIM IMAGER* software supports export of acquired measurement data for further analysis and storage. Users need to define define export file name and storage locations to export measurement datasets. Additionally, users can choose to add metadata and experiment notes which are stored as .xml files. Detailed information about export formats and post-processing tools is provided in the *FLIM IMAGER* software manual.

**Notice** Verify that data export is enabled as indicated on the Export button before starting long acquisitions to avoid unintended data loss.

## 5.6 Shutdown Procedure

After completing measurements:

1. Stop FLIM acquisition in the *FLIM IMAGER* software.
2. Stop microscope scanning.
3. Save and export all acquired measurement data.
4. Close the *FLIM IMAGER* software.
5. Close the openUC2 control software.
6. Power down all FLIM LABS hardware components.
7. Power down the microscope system according to the microscope manufacturer's instructions.

## 5.7 Troubleshooting

| Issue | Possible Cause | Solution |
|---|---|---|
| No FLIM LABS hardware detected | USB connection or driver issue. | Check the USB connection, restart the software, and reinstall the driver if required. |
| No synchronization signal detected | Incorrect synchronization connection or trigger configuration. | Check the synchronization signal, cable connections, and trigger settings. |
| No photon counts detected | Detector signal missing or incorrect input selected. | Check detector connections, input channel selection, and acquisition settings. |
| Distorted or unstable lifetime result | Incorrect timing, excessive count rate, or unstable synchronization. | Reduce the count rate, verify synchronization, and review acquisition settings. |
| Unstable acquisition | USB bandwidth or connection issue. | Use a direct USB connection and avoid unsupported USB hubs. |

# 6 Maintenance and Service

This chapter describes the cleaning, preventive maintenance, corrective maintenance, and authorized service procedures for the supplied hardware components. All maintenance activities must be performed by qualified personnel who have read and understood this manual.

Maintenance must be carried out in accordance with this manual and applicable workplace safety regulations. The maintenance area should provide adequate illumination; a minimum lighting level of 200 lux is recommended.



Before performing any cleaning or maintenance operation, switch off all powered components and disconnect them from their power sources.

## 6.1 Cleaning

The system components require periodic external cleaning to ensure reliable operation.

- Remove dust or dirt from exterior surfaces using a dry or slightly damp lint-free cloth.
- A mild, neutral detergent may be used if necessary.
- Do not allow liquids to enter connectors, optical interfaces, or electronic components.
- Avoid the use of solvents, aggressive chemicals, or abrasive materials.

For connectors and hard-to-reach areas, such as USB ports or ventilation openings, a soft-bristled brush or compressed air may be used.

### Optical Port and Connector Cleaning

Proper cleaning of optical connectors is essential to ensure reliable optical coupling and to prevent signal attenuation or permanent damage to optical interfaces.

Always inspect the ferrule end faces of fiber patch cables and optical bulkhead connectors before making optical connections. Ferrule end faces may be cleaned using commercially available lint-free fiber cleaning wipes or by applying a small amount of isopropanol (IPA) to a clean lint-free cloth.

Optical bulkhead connectors are best cleaned using a dedicated fiber-optic cleaning pen or cleaning stick designed for the connector type in use. Follow the instructions provided by the manufacturer of the cleaning tool.



**Caution** Before cleaning any optical connector or fiber interface, ensure that all laser sources are switched off and that no optical radiation is present in the system.

## 6.2 Preventive Maintenance

The supplied FLIM LABS hardware components do not require routine internal maintenance.

Periodically inspect:

- External surfaces for mechanical damage
- Connectors and cables for wear, deformation, or contamination
- Optical connectors and fiber patch cables for cleanliness and damage
- Product labels for legibility

If any abnormal condition is detected, discontinue use of the affected component and contact technical support.

## 6.3 Service and Repairs

The supplied hardware components contain no user-serviceable internal parts. In the event of malfunction, accidental damage, or improper operation:

- Switch off and disconnect the affected component.
- Do not attempt internal repair.

- Contact the manufacturer’s technical support for instructions.

Repairs or modifications must be carried out exclusively by FLIM LABS S.r.l. or authorized personnel. Use only original FLIM LABS spare parts and approved replacement accessories. Further warranty conditions are described in Section 1.6.



Unauthorized intervention or the use of non-original components may compromise safety, performance, and warranty coverage.

## 6.4 Firmware and Software Updates

Firmware and software updates may be provided to improve performance, compatibility, or reliability. Use only firmware and software updates supplied or approved by the manufacturer. Follow the update instructions provided with the update package or in the dedicated software documentation.

Do not disconnect power or data cables during firmware updates unless explicitly instructed to do so. Interrupted updates may cause malfunction or require service intervention.

# 7 Decommissioning and Disposal

This chapter describes procedures for removing system components from service and disposing of them in accordance with applicable regulations.

## 7.1 Decommissioning Procedure

The supplied hardware components are designed for long-term laboratory use. When a component reaches the end of its technical or operational life, it must be decommissioned to prevent further use.

Decommissioning requires:

- Disconnecting the component from all power sources.
- Removing all signal, communication, and optical connections.
- Ensuring that the component cannot be unintentionally reused.

The same procedure applies in cases of:

- Permanent withdrawal from service
- Storage without intended future use
- Final disposal

Reuse of components after final decommissioning is not permitted unless explicitly authorized by the manufacturer.

## 7.2 Disposal

All hardware components supplied with the system must be disposed of in accordance with applicable local and European regulations governing electrical and electronic equipment.

In accordance with Directive 2012/19/EU (WEEE), electronic components must not be disposed of with household waste. The product labels include the WEEE symbol indicating the requirement for separate collection of electrical and electronic equipment. Contact your local waste management authority or a certified collection center for proper disposal procedures.

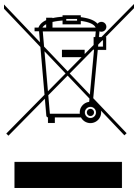


**Figure 7.1:** WEEE symbol